

The Kannan–Lovász–Simonovits Conjecture

Problem, Audited Progress, and Possible Future Directions

Research note from a multi-route proof investigation

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Abstract

The Kannan–Lovász–Simonovits (KLS) conjecture asks for a dimension-free Cheeger, or equivalently Poincaré, inequality for every isotropic log-concave probability measure. This note gives a standalone statement of the problem, records the exact reductions needed to pass between its standard formulations, and summarizes the strongest results obtained in an extended proof investigation. The main advances are an exact second-order ledger for a low eigenfunction, a rank-one-defect Lichnerowicz theorem, strict spectral amplification under normalized self-convolution, including its unequal-input extension, and a posterior reverse-smoothing theorem which makes fixed-Gaussian analytic outputs a quantitatively equivalent target. A weighted-Fisher ledger closes for pure translations and for smooth post-noise Gaussian conditional fibers. Exact stochastic-localization and Gaussian-posterior identities isolate the remaining global gates. We also record explicit counterexamples to several tempting forward-only and slice-local claims. No complete proof of KLS is claimed. Checkpoint 17 adds audited posterior Hilbert–Schmidt and affine-stability ledgers, the asymptotic fixed-Gaussian saturation comparison, simple-eigenspace and longitudinal Hardy frontiers, and the product ANOVA/matrix-Stein obstruction. The final section proposes a prioritized research program whose statements are precise enough to be proved or falsified independently.

Status. Checkpoint 17 does not contain a complete proof of KLS. Every assertion labeled “open target,” “candidate,” or “future direction” remains unproved. The posterior reverse theorem is a valid general reduction, while the score and weighted-Fisher estimates close only near-affine branches. The affine-interaction inequality needed for the remaining nonlinear branch is not proved. All other displayed identities and model calculations below were checked with their dependence on dimension and auxiliary parameters exposed. The new posterior Hilbert–Schmidt, affine-stability, fixed-Gaussian, simple-eigengap, and product-ANOVA statements are audited lemmas or conditional frontiers. The twice-Bochner ledger still retains an $O_\kappa(\lambda)$ transverse saturation remainder and a longitudinal quotient gate; neither is assumed away. These results are not a KLS proof.

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1 The problem

1.1 Log-concavity, boundary measure, and covariance

A probability measure μ on $\mathbb{R}^n$ is *log-concave* if for compact sets $A, B \subset \mathbb{R}^n$ and $0 < \theta < 1$,

$$\mu(\theta A + (1 - \theta)B) \geq \mu(A)^\theta \mu(B)^{1-\theta}.$$

By Borell's characterization, μ is supported on an affine subspace E ; on its minimal affine hull it is either a point mass or has a density e^{-V} , where V is convex [1].

For a Borel set A , let

$$A_\varepsilon = \{x : \text{dist}(x, A) \leq \varepsilon\}, \quad \mu^+(A) = \liminf_{\varepsilon \downarrow 0} \frac{\mu(A_\varepsilon) - \mu(A)}{\varepsilon}.$$

The Cheeger constant is

$$\psi_\mu = \inf_{0 < \mu(A) < 1} \frac{\mu^+(A)}{\min\{\mu(A), 1 - \mu(A)\}}.$$

Write $A_\mu = \text{Cov}(\mu)$. The measure is *isotropic* when its barycenter is zero and $A_\mu = I$.

1.2 The KLS conjecture

Theorem 1.1 (KLS conjecture; desired statement). *There is a universal constant $c > 0$ such that every non-point log-concave probability measure μ on every Euclidean space satisfies*

$$\boxed{\psi_\mu \geq \frac{c}{\sqrt{\|A_\mu\|_{\text{op}}}}.} \tag{KLS}$$

Equivalently, $\inf_n \inf_\mu \psi_\mu > 0$, where the second infimum is over isotropic log-concave probabilities on $\mathbb{R}^n$.

The conjecture originated in the study of random walks and isoperimetry for convex bodies [2]. The word *universal* is essential: no dependence on n , the support, a smoothing parameter, or a localization time is permitted.

2 Exact reductions and equivalent targets

2.1 Affine hull and degenerate covariance

Let $E = m + L$ be the minimal affine hull of the support. Exterior Minkowski content computed in $\mathbb{R}^n$ agrees with that computed using the induced Euclidean metric on E . Indeed, for $B = A \cap E$, $B_\varepsilon^E \subset A_\varepsilon \cap E$, while a set already contained in E has the same distance function on E in either ambient space. Taking the two infima gives

$$\psi_\mu^{\mathbb{R}^n} = \psi_\mu^E. \quad (2.1)$$

On the direction space L , the covariance is positive definite. If $v \in L$ had zero variance, the support would lie in the proper affine hyperplane $m + (L \cap v^\perp)$, contradicting minimality. On $L^\perp$, covariance is zero. Thus the ambient operator norm is the largest eigenvalue of $A_\mu|_L$.

2.2 Affine maps

If $S : E \rightarrow F$ is invertible and $\nu = S_\# \mu$, then

$$\frac{\psi_\mu}{\|S\|_{\text{op}}} \leq \psi_\nu \leq \|S^{-1}\|_{\text{op}} \psi_\mu. \quad (2.2)$$

For example, $A_{\varepsilon/\|S\|_{\text{op}}} \subset S^{-1}((SA)_\varepsilon)$, which proves the left inequality directly from the definition; apply it to S^{-1} for the other direction.

Translate by $-m$ and take $S = A_\mu^{-1/2}$ on L . The image is isotropic, and $\|S^{-1}\|_{\text{op}} = \sqrt{\|A_\mu\|_{\text{op}}}$. Consequently, a universal isotropic lower bound implies (KLS), including all lower-dimensional supports. The converse is immediate. A point mass is precisely the excluded zero-dimensional case.

2.3 Poincaré and first-moment formulations

Let $C_P(\mu)$ be the optimal constant in

$$\text{Var}_\mu f \leq C_P(\mu) \int |\nabla f|^2 d\mu. \quad (2.3)$$

The chain rule gives

$$C_P(S_\# \mu) \leq \|S\|_{\text{op}}^2 C_P(\mu), \quad C_P(\mu) \leq \|S^{-1}\|_{\text{op}}^2 C_P(S_\# \mu). \quad (2.4)$$

For log-concave measures, Cheeger's inequality and the reverse Buser–Ledoux implication under $CD(0, \infty)$, together with standard convex approximation, give universal constants $c, C > 0$ such that

$$C_P(\mu) \leq \frac{4}{\psi_\mu^2}, \quad \psi_\mu \geq \frac{c}{\sqrt{C_P(\mu)}}. \quad (2.5)$$

Hence KLS is equivalent, up to constants, to

$$\boxed{\text{Var}_\mu f \leq C \|A_\mu\|_{\text{op}} \int |\nabla f|^2 d\mu} \quad (\text{P})$$

for every locally Lipschitz $f \in L^2(\mu)$. Truncation, cutoff, and mollification extend a proof for compactly supported smooth tests to this full class without changing the constant.

E. Milman's equivalence theorem under the same convexity hypothesis identifies ψ_μ^{-1} , up to constants, with

$$D_1(\mu) = \sup_{\text{Lip}(f) \leq 1} \int |f - \text{med}_\mu f| d\mu [4].$$

Mean and median centerings differ by at most a factor two. Thus the isotropic form of KLS is also equivalent to

$$\sup_{\text{Lip}(f) \leq 1} \mathbb{E}|f(X) - \mathbb{E}f(X)| \leq C. \quad (\text{M})$$

Linear functions already satisfy this by isotropy. Radial Lipschitz functions are controlled by the dimension-free thin-shell theorem. A bad witness must therefore be genuinely nonlinear and non-radial.

Remark 2.1. KLS is not a dimension-free log-Sobolev statement. Isotropic one-dimensional exponential measures already obstruct subgaussian tails. No argument below assumes a uniform log-Sobolev or T_2 inequality.

3 An exact low-eigenfunction ledger

This section records the most useful general identities found in the investigation. Work first with a smooth convex potential on a smooth convex support and Neumann boundary condition; the closed-form identities extend by convex approximation.

Let $A = -L_\mu$ be the nonnegative weighted generator and suppose

$$Af = \lambda f, \quad \mathbb{E}f = 0, \quad \mathbb{E}f^2 = 1, \quad \mathbb{E}|\nabla f|^2 = \lambda.$$

Set

$$a = \mathbb{E}[Xf], \quad r = f - a \cdot x, \quad \delta = \|r\|_2.$$

Proposition 3.1 (Affine residual identities). *For an isotropic μ ,*

$$\mathbb{E}r = 0, \quad \mathbb{E}[Xr] = 0, \quad |a|^2 = 1 - \delta^2, \quad (3.1)$$

$$\mathbb{E}\nabla f = \lambda a, \quad \mathbb{E}\nabla r = -(1 - \lambda)a, \quad (3.2)$$

$$\mathbb{E}|\nabla r|^2 = 1 - \lambda + (2\lambda - 1)\delta^2, \quad (3.3)$$

$$\text{Var}(\nabla f) = \lambda[1 - \lambda(1 - \delta^2)]. \quad (3.4)$$

Proof. The first line is orthogonal projection onto the linear functions. Testing the weak eigenvalue equation against $b \cdot x$ gives $b \cdot \mathbb{E}\nabla f = \lambda b \cdot a$, proving (3.2). The remaining formulas follow by expanding $\nabla r = \nabla f - a$. $\square$

Let

$$B_f = \mathbb{E} \left\| D^2 f \right\|_{\text{HS}}^2$$

and let $\mathcal{R}_V(f) \geq 0$ denote the sum of the potential-curvature and Reilly boundary-curvature terms. Bochner–Reilly gives

$$B_f + \mathcal{R}_V(f) = \lambda^2. \quad (3.5)$$

Applying the same bottom Poincaré inequality componentwise to $\partial_i f$ and using (3.4) produces the exact nonnegative defect identity

$$\boxed{[B_f - \lambda \text{Var}(\nabla f)] + \mathcal{R}_V(f) = \lambda^3(1 - \delta^2).} \quad (3.6)$$

In particular,

$$\|D^2 r\|_2 \leq \lambda. \quad (3.7)$$

The identity says that a hypothetical small-gap, near-linear eigenfunction has a residual r that is small both in value and Hessian, yet has mean gradient almost $-a$. This isolates the following gate.

Open target 3.2 (Residual mean-gradient inequality). Prove, for every centered isotropic log-concave μ ,

$$\boxed{|\mathbb{E}\nabla g| \leq C(\|g\|_2 + \|D^2 g\|_2), \quad \mathbb{E}g = 0, \quad \mathbb{E}[Xg] = 0.} \quad (\text{MG})$$

Applied to r , (MG) gives

$$(1 - \lambda)\sqrt{1 - \delta^2} \leq C(\delta + \lambda), \quad (3.8)$$

and therefore closes the near-linear branch whenever δ is a sufficiently small universal constant.

The stronger Hessian-only form holds in one dimension and tensorizes to products. If ν is centered, variance one, and log-concave on $\mathbb{R}$, its one-dimensional Poincaré constant is universally bounded. Applying Poincaré first to h' and then to $h - (\int h' d\nu)x$ gives, for $h \perp \{1, x\}$,

$$\left| \int h' d\nu \right| \leq 12 \left(\int (h'')^2 d\nu \right)^{1/2}. \quad (3.9)$$

Conditional projections onto product coordinates are orthogonal, so

$$|\mathbb{E}\nabla g|^2 \leq 144 \mathbb{E} \|D^2 g\|_{\text{HS}}^2 \quad (3.10)$$

for products of isotropic one-dimensional log-concave laws.

3.1 A false interpolation and its diagnostic value

The seemingly natural inequality

$$\text{Var}(\nabla f) \leq C\delta \|D^2 f\|_2 \quad (3.11)$$

is false even for genuine first eigenfunctions. The first degree-one Neumann mode on the isotropic Euclidean ball has

$$\delta \asymp n^{-1}, \quad \text{Var}(\nabla f) \asymp n^{-1}, \quad \|D^2 f\|_2 \asymp n^{-1/2},$$

so the ratio in (3.11) grows like $\sqrt{n}$. Any viable near-linearity argument must target the *mean* gradient or another finite-dimensional component, not the whole gradient variance.

4 The directional third-moment gate

For a symmetric matrix B , define

$$q_B = X^T B X - \text{Tr } B, \quad T B = \mathbb{E}[X q_B].$$

The quadratic restriction of the Hessian-only form of (MG) is exactly

$$\sup_{|u|=1} \left\| \mathbb{E}[(u \cdot X)(X X^T - I)] \right\|_{\text{HS}} \leq C. \quad (\text{TM})$$

Indeed, $T^* u = \mathbb{E}[(u \cdot X)(X X^T - I)]$, while $D^2 q_B = 2B$. More explicitly, apply the inequality to $q_B - (TB) \cdot X$, which is orthogonal to both constants and linear functions and has the same Hessian as q_B .

Thin-shell estimates for every projection give only

$$|\text{Tr}(P M_u)| \leq C \sqrt{\text{rank } P}, \quad M_u = \mathbb{E}[(u \cdot X)(X X^T - I)]. \quad (4.1)$$

Ordering the eigenvalues yields $|\lambda_j(M_u)| \leq C j^{-1/2}$, hence

$$\|M_u\|_{\text{HS}} \leq C \sqrt{\log(en)}. \quad (4.2)$$

The harmonic diagonal matrix shows that (4.1) alone cannot remove the logarithm. A successful proof of (TM) must use compatibility among matrix directions, not only rank-by-rank trace information.

4.1 Dependent model classes where (TM) is dimension-free

The investigation established (TM) with universal constants for several nonproduct classes.

Paired wedges. For

$$K_a = \{(t, x, y) : |t| \leq 1, |x_i| \leq 1 + a_i t, |y_i| \leq 1 - a_i t\},$$

let $S = \sum_i a_i^2$ and $v = \text{Var } T$. The axial marginal potential satisfies $W'' \geq 2S$, hence $vS \leq 1/2$. After whitening, the only nonzero third coefficients are

$$\lambda_i = \frac{2a_i \sqrt{v}}{1 + a_i^2 v}, \quad \sum_i \lambda_i^2 \leq 2,$$

and therefore

$$\sup_{|u|=1} \|M_u\|_{\text{HS}} \leq 2. \quad (4.3)$$

The whitened law is an explicit 4-Lipschitz image of a product law, which also gives $C_P \leq 192$. A one-sided wedge admits an analogous map and the explicit bound $C_P < 972$.

Dirichlet laws. For a log-concave Dirichlet distribution with parameters $\alpha_i \geq 1$, write $A = \sum_i \alpha_i$ and $q_i = \alpha_i/A$. A unit tangent direction has coefficients a_i with $\sum q_i a_i = 0$ and $\sum q_i a_i^2 = 1$. Direct moment algebra gives

$$\|M_{u(a)}\|_{\text{HS}}^2 = \frac{4(A+1)}{(A+2)^2} \left(\sum_i a_i^2 - 2 \right) < 4. \quad (4.4)$$

Cones and conditional Gaussian families. For a cone over a centrally symmetric base,

$$\sup_{|u|=1} \|M_u\|_{\text{HS}}^2 = \frac{4(n-1)(n+2)}{(n+3)^2} < 4. \quad (4.5)$$

The same gate holds for affine box slices and affine Gaussian covariance pencils.

More generally, consider

$$p(t, y) \propto \exp \left[-W(t) - \frac{1}{2} y^T Q(t) y \right], \quad R(t) = Q(t)^{-1}.$$

Joint log-concavity is equivalent to

$$W'' \geq 0, \quad R'' \preceq 0. \quad (4.6)$$

This permits genuinely rotating, noncommuting covariance eigenspaces. After whitening, marginal curvature controls the normalized covariance velocity by

$$\Phi''(t) \geq \frac{1}{2} \left\| R(t)^{-1/2} R'(t) R(t)^{-1/2} \right\|_{\text{HS}}^2. \quad (4.7)$$

A one-dimensional Stein-kernel argument then proves (TM) with a universal constant for the entire conditional-Gaussian class.

4.2 Why general slicing remains open

For an arbitrary smooth conditional law $p_t(y) \propto e^{-V(t,y)}$, set $H = D_y^2 V$, $g = V_t - \mathbb{E}_t V_t$, and let $Lu = g$ for the conditional generator. The Brascamp–Lieb deficit controls $\mathbb{E}_t \|D^2 u\|_{\text{HS}}^2$. Covariance velocity would follow from

$$\boxed{\|\mathbb{E}[Y \otimes \nabla u + \nabla u \otimes Y]\|_{\text{HS}}^2 \leq C \mathbb{E} \|D^2 u\|_{\text{HS}}^2.} \quad (\text{AH})$$

This affine-Hessian rigidity is itself a near-linear KLS gate. Moving boundaries cannot be omitted: for an interval and for a centered exponential law, the covariance of the iid sum-difference slice is convex, not concave, and all relevant curvature is carried by endpoint flux.

5 A rank-one-defect Lichnerowicz theorem

One genuinely general positive theorem arose from the localization route.

Proposition 5.1 (Rank-one curvature defect). *Let $u \in S^{n-1}$, $P = I - u \otimes u$, and let $d\mu = e^{-V} dx$ be log-concave. If, distributionally,*

$$D^2 V \succeq \kappa P,$$

then

$$\boxed{C_P(\mu) \leq 201 \left(\kappa^{-1} + \text{Var}_\mu \langle X, u \rangle \right).} \quad (5.1)$$

The same statement holds intrinsically on a lower-dimensional affine support.

Here is the proof mechanism. Disintegrate at $x = tu + y$, with conditional law ν_t on $u^\perp$ and marginal $e^{-U(t)}dt$. There is a canonical continuity-equation velocity w_t satisfying

$$\partial_t \nu_t + \operatorname{div}(\nu_t w_t) = 0, \quad \kappa \int |w_t|^2 d\nu_t \leq U''(t). \quad (5.2)$$

The second inequality is a reinforced infinitesimal Prékopa theorem; the cross-Hessian block is essential. If $g(t) = \int f(t, y) d\nu_t(y)$, then

$$g'(t) = \int (\partial_t f + \nabla_y f \cdot w_t) d\nu_t,$$

so

$$\frac{g'(t)^2}{1 + \int |w_t|^2 d\nu_t} \leq \int |\nabla f|^2 d\nu_t. \quad (5.3)$$

A curvature-weighted one-dimensional Hardy inequality, plus conditional Lichnerowicz in $u^\perp$, proves (5.1). Gaussian convolution preserves the transverse curvature quantitatively:

$$\kappa \mapsto \frac{\kappa}{1 + \kappa \varepsilon},$$

which permits passage to nonsmooth potentials without loss in the final constant.

This theorem is useful after localization has created curvature in all but one direction. Its remaining obstruction is the variance of the adaptively selected survivor direction.

6 Stochastic localization and heat-flow identities

6.1 Function-specific localization

For a normalized first eigenfunction, a localization driver can preserve its posterior mean while adding curvature away from one selected direction. Combining the preserved variance with (5.1) gives the exact ledger

$$1 \leq \frac{192}{T} \lambda + 96 \mathbb{E}[R_T \mathcal{E}_T]. \quad (6.1)$$

Here R_T is the terminal variance in the surviving direction and $\mathcal{E}_T = \mathbb{E}_T |\nabla f|^2$ is posterior energy. Thus a small gap forces R_T to be of order $1/\lambda$ under the energy-biased path law. Ordinary expectation bounds on R_T are insufficient because of this correlation.

A softened driver that creates full curvature has the complementary tradeoff

$$1 - \mathbb{E} m_T^2 \leq \frac{\lambda}{\eta T}, \quad (6.2)$$

where m_T is the localized mean and $\eta T I$ is the curvature created. Equations (6.1)–(6.2) make precise why preserving the signal and regularizing its selected direction compete.

6.2 Parallel coupling and a low-mode martingale

In standard stochastic localization, write $A_t = \operatorname{Cov}(\mu_t)$ and let

$$M_0 = I, \quad \dot{M}_t = A_t M_t.$$

For a fixed f , set

$$c_t = \text{Cov}_{\mu_t}(X, f), \quad D_t = \mathbb{E}_t[(f - \mathbb{E}_t f)(X - a_t)(X - a_t)^T].$$

Direct Itô differentiation gives

$$dc_t = D_t dB_t - A_t c_t dt, \quad \boxed{M_t^T c_t \text{ is a martingale.}} \quad (6.3)$$

For the eigenfunction decomposition in [Section 7](#) below, $c_0 = a$. At time one, μ_1 is 1-strongly log-concave, so

$$\mathbb{E}|c_1|^2 \leq \lambda. \quad (6.4)$$

With $b = a/|a|$, (6.3)–(6.4) imply

$$\boxed{\mathbb{E}|M_1 b|^2 \geq \frac{1 - \delta^2}{\lambda}.} \quad (6.5)$$

A nearly linear small mode must therefore select an expensive direction of the coupling. Available theorems control only the trace

$$\mathbb{E} \text{Tr}(M_1^T M_1) \leq Cn, \quad (6.6)$$

not the selected direction.

The weakest final-time statement that closes this branch is

$$\boxed{\mathbb{E} \left[\frac{\langle M_1 b, c_1 \rangle^2}{|c_1|^2} \mathbf{1}_{\{c_1 \neq 0\}} \right] \leq C.} \quad (\text{FA})$$

This is strictly weaker than $\mathbb{E}|M_1 b|^2 \leq C$: it controls only the random normal component used by the martingale pairing. It passes the Gaussian, cube, product-exponential, ball, and simplex tests. A one-spike matrix path shows that (6.6) and ordered-rank stopping bounds alone do not imply (FA).

Coarea does not automatically help. The quotient in (FA) is a normal stretch of the initial-tilt flow at one fixed starting point. Change of variables averages over random preimages instead, and endpoint integration by parts controls the conditional mean of $M_1 b$, not its conditional second moment.

7 Gaussian posterior and a directional-MMSE target

Let $X \sim \mu$, $G \sim N(0, I)$ be independent, and $Y = X + G$. Write

$$m(y) = \mathbb{E}[X \mid Y = y], \quad A(y) = \text{Cov}(X \mid Y = y).$$

The posterior is 1-strongly log-concave, hence $0 \preceq A(y) \preceq I$. For $F(y) = \mathbb{E}[f(X) \mid Y = y]$, exponential-family differentiation gives

$$\nabla F(y) = \text{Cov}(X, f \mid Y = y) = c(y). \quad (7.1)$$

If $f = a \cdot x + r$, then exactly

$$c(y) = A(y)a + e(y), \quad e(y) = \text{Cov}(X, r \mid Y = y). \quad (7.2)$$

Posterior Poincaré and total variance imply

$$\mathbb{E}|c(Y)|^2 \leq \lambda, \quad \mathbb{E}|e(Y)|^2 \leq \delta^2. \quad (7.3)$$

For a unit vector b , define

$$\text{mmse}_\mu(b) = \mathbb{E} \text{Var}(b \cdot X \mid X + G) = \mathbb{E}[b^T A(Y) b]. \quad (7.4)$$

Open target 7.1 (Directional MMSE floor). Prove that every centered isotropic log-concave law satisfies

$$\inf_{|b|=1} \text{mmse}_\mu(b) \geq c_0 > 0. \quad (\text{DMMSE})$$

This would close the very-near-linear branch immediately. Indeed, with $a = |a|b$, (7.2)–(7.3) and the Hilbert-space triangle inequality give

$$\sqrt{\lambda} \geq c_0 \sqrt{1 - \delta^2} - \delta. \quad (7.5)$$

The target is nonlinear despite involving a linear latent variable. The orthogonal noisy coordinates may carry nonlinear information about $b \cdot X$. Tweedie's identity

$$m(y) = y + \nabla \log q(y), \quad \nabla m(y) = A(y), \quad (7.6)$$

where q is the density of Y , shows the KLS content. For $h_b(y) = b \cdot m(y)$,

$$\text{Var}_q(h_b) = 1 - \text{mmse}_\mu(b), \quad \int |\nabla h_b|^2 dq = \mathbb{E}|A(Y)b|^2. \quad (7.7)$$

Thus failure of (DMMSE) produces a low-energy monotone posterior-mean function for the log-concave Gaussian convolution q .

There is a possible dimension-descent estimate worth auditing. Reveal $Y_\perp = P_{b^\perp} Y$ first and put

$$v(z) = \text{Var}(b \cdot X \mid Y_\perp = z), \quad h_0(z) = \mathbb{E}[b \cdot X \mid Y_\perp = z].$$

The rank-one curvature defect gives transverse posterior covariance at most I , hence

$$|\nabla h_0(z)|^2 \leq v(z). \quad (7.8)$$

Conditional one-dimensional log-concavity gives a scalar Gaussian-channel MMSE comparable from below to $\min(v, 1)$. One-dimensional log-concave moments suggest $\|v\|_p \lesssim p^2$. Combining these facts with Poincaré for the $(n-1)$ -dimensional marginal suggests the candidate recursion

$$\text{mmse}_\mu(b) \gtrsim \frac{1}{C_P(q_\perp) \log^2(e + C_P(q_\perp))}. \quad (7.9)$$

Equation (7.9) was not fully audited before the investigation stopped. It would still contain logarithmic loss, but it offers a concrete starting point for a new dimension-descent mechanism.

8 Further exact progress

8.1 Convex half-set distance

If A is convex and $\mu(A) = 1/2$, one-dimensional localization of parallel expansions gives the dimension-free lower bound

$$\mathbb{E} \operatorname{dist}(X, A) \geq \frac{2 \log 2 - 1}{4\mu^+(A)}. \quad (8.1)$$

The scalar constant is explicit. The reverse inequality for arbitrary half-sets is already KLS-strength, and naive convexification fails even for the isotropic Laplace law.

8.2 Tensor slow-growth plateaus

Let

$$\mathcal{C}_n = \sup\{C_P(\mu) : \mu \text{ isotropic log-concave on } \mathbb{R}^n\}.$$

If $\mathcal{C}_n$ were unbounded while obeying a logarithmic upper envelope, monotonicity implies dimensions n_j and integers $r_j \rightarrow \infty$ such that

$$\frac{\mathcal{C}_{r_j n_j}}{\mathcal{C}_{n_j}} \rightarrow 1. \quad (8.2)$$

Thus large tensor powers are relative near-extremizers with growing first eigenspace multiplicity.

This does not by itself create a contradiction. For a mixed tensor mode indexed by a symmetric zero-diagonal matrix B , the spectral splitting is at most the convex-buffer cost times

$$\frac{\|B\|_{\text{op}}}{\|B\|_{\text{HS}}} \leq 1. \quad (8.3)$$

Moreover, the r tensor low modes occupy a rank- r exceptional subspace in ambient dimension rn ; ordered-rank localization still sees $\log((rn)/r) = \log n$. Multiplicity alone therefore does not remove the small-rank obstruction.

9 Self-convolution and the weighted-slice program

9.1 Strict forward amplification under self-convolution

Let X, Y be independent with the same centered isotropic log-concave law μ , and put

$$S = \frac{X+Y}{\sqrt{2}}, \quad \nu = \mathcal{L}(S), \quad a = \lambda_1(\mu) = C_P(\mu)^{-1}, \quad b = \lambda_1(\nu) = C_P(\nu)^{-1}.$$

Proposition 9.1 (Forward self-convolution amplification). *Without assuming that either spectral edge is attained,*

$$b(1-b) \leq 4(b-a), \quad b \geq \frac{\sqrt{9+16a}-3}{2}. \quad (\text{SC})$$

For a centered unit test $f(S)$, set

$$h(X) = \mathbb{E}[f(S) | X], \quad R_f = f(S) - h(X) - h(Y), \quad q = \mathbb{E}|\nabla f(S)|^2.$$

The product Poincaré inequality on the two-coordinate-centered residual and the vector Hoeffding decomposition give

$$E_R := \mathbb{E}|\nabla_{X,Y} R_f|^2 \leq 2(q - a), \quad (10.1)$$

$$\text{Var}(Z) = 2 \text{Var}(U) + \mathbb{E}|W|^2, \quad E_R = \text{Var}(U) + \mathbb{E}|W|^2, \quad (10.2)$$

where $Z = \nabla f(S)$, $U = \mathbb{E}[Z | X]$, and $W = Z - \mathbb{E}[Z | X] - \mathbb{E}[Z | Y] + \mathbb{E}Z$. Thus $\text{Var}(\nabla f) \leq 2E_R \leq 4(q - a)$. Taking f in a spectral window $[b, b + \varepsilon]$, testing the weak generator equation against coordinates, and sending $\varepsilon \downarrow 0$ gives $\text{Var}(\nabla f) \geq b(1 - b) - o(1)$. This proves (SC), including at a continuous bottom spectral edge.

If a_k is the gap after k normalized dyadic convolutions, then $a_k \uparrow 1$, and

$$a_{k+1} \leq \frac{1}{2} \implies a_{k+1} \geq \frac{8}{7}a_k. \quad (10.3)$$

This is only forward amplification. It may take $O(\log(1/a_0))$ convolutions to reach a fixed gap, so (SC) gives no dimension-free lower bound on the initial a_0 .

9.2 Unequal inputs and fixed-Gaussian output structure

Equality of the two input laws is not needed. Let μ_0, μ_1 be centered isotropic log-concave laws on the same Euclidean space, let $X \sim \mu_0$, $Y \sim \mu_1$ be independent, and set

$$S = \frac{X + Y}{\sqrt{2}}, \quad a = \min\{\lambda_1(\mu_0), \lambda_1(\mu_1)\}, \quad b = \lambda_1(\mathcal{L}(S)).$$

The unequal vector predictors need not have the same variance. If

$$u = \mathbb{E}|\mathbb{E}[\nabla f(S) | X] - \mathbb{E}\nabla f(S)|^2, \quad v = \mathbb{E}|\mathbb{E}[\nabla f(S) | Y] - \mathbb{E}\nabla f(S)|^2$$

and w is the squared norm of the two-coordinate residual, then

$$\text{Var}(\nabla f) = u + v + w, \quad E_R = \frac{u + v}{2} + w.$$

Consequently the proof of (SC) is unchanged:

$$b(1 - b) \leq 4(b - a), \quad b \geq \frac{\sqrt{9 + 16a} - 3}{2}.$$

(USC)

The identities are first read for smooth tests and then in the closed weighted Sobolev form domain. For log-concave laws, smooth tests form a core after working intrinsically on the minimal affine hull. This note does not extend (USC) to arbitrary singular non-log-concave inputs.

Now take $\mu_1 = \gamma = N(0, I)$ and define

$$\mathcal{G}\mu = \mathcal{L}\left(\frac{X + G}{\sqrt{2}}\right).$$

Since every isotropic law has $\lambda_1(\mu) \leq 1 = \lambda_1(\gamma)$, the input parameter in (USC) is $a = \lambda_1(\mu)$. The only rearrangement supplied by (USC) is

$$\lambda_1(\mu) \leq \frac{\lambda_1(\mathcal{G}\mu)(3 + \lambda_1(\mathcal{G}\mu))}{4}. \quad (10.3a)$$

This forward constraint gives no lower bound on the input from a lower bound on the output. Hence (USC) alone gives no fixed-Gaussian reduction of KLS. There is, however, a separate reverse comparison.

Proposition 9.2 (Posterior reverse smoothing). *Let $a = \lambda_1(\mu)$ and $b = \lambda_1(\mathcal{G}\mu)$. Then*

$$\boxed{b \leq \frac{2a}{1-a} \quad (a < 1), \quad a \geq \frac{b}{2+b}.} \quad (10.3c)$$

The second inequality also holds when $a = 1$.

Indeed, take a centered normalized input form-domain test f with energy $q < 1$, put $Y = X + G$, and set $F(y) = \mathbb{E}[f(X) \mid Y = y]$. The posterior is intrinsically one-strongly log-concave, so

$$\mathbb{E} \text{Var}(f(X) \mid Y) \leq q, \quad \text{Var}(F(Y)) \geq 1 - q. \quad (10.3d)$$

Posterior differentiation gives $\nabla F = \text{Cov}(X, f \mid Y)$. Covariance Cauchy–Schwarz together with $\text{Cov}(X \mid Y) \preceq I$ gives the dimension-free pointwise estimate

$$|\nabla F(Y)|^2 \leq \text{Var}(f(X) \mid Y),$$

and hence the energy of $s \mapsto F(\sqrt{2}s)$ under $\mathcal{G}\mu$ is at most $2q$. The variational definition of b , followed by a form-domain minimizing sequence $q \downarrow a$, proves (10.3c). The argument is intrinsic on a lower-dimensional affine support and uses posterior strong log-concavity for hard supports, so no spectral attainment or ambient full-dimensionality is assumed.

The output has a positive analytic log-concave density. Before the $1/\sqrt{2}$ rescaling, if $U = -\log(\mu * \gamma)$, the Gaussian posterior identity and posterior Brascamp–Lieb inequality give

$$D^2U(y) = I - \text{Cov}(X \mid X + G = y), \quad 0 \preceq \text{Cov}(X \mid X + G = y) \preceq I.$$

For the potential $\tilde{U}(s) = U(\sqrt{2}s) + \text{const}$ of $\mathcal{G}\mu$,

$$0 \preceq D^2\tilde{U} \preceq 2I. \quad (10.3b)$$

The forward inequality (USC) does not reverse, but the independent posterior estimate (10.3c) does. Therefore the positive analytic isotropic class

$$0 \preceq D^2\tilde{U} \preceq 2I$$

is a quantitatively equivalent KLS target: a universal output gap $b \geq c_0$ gives $a \geq c_0/(2 + c_0)$, while the converse direction is restriction to a subclass. This upper-curvature condition is not a Bakry–Émery lower-curvature hypothesis. Lower-dimensional inputs and their Gaussian noises are handled intrinsically on their affine hulls.

9.3 Score rigidity, affine interaction, and one-form circularity

On the regularized class $0 \preceq D^2V \preceq \beta I$, integration by parts and covariance positivity give the score-rigidity estimate

$$|\mathbb{E}\nabla g| \leq \sqrt{\beta - 1} \|g\|_2 \quad (10.3e)$$

for every affine-orthogonal g . At $\beta = 2$, this gives a numerical gap for every bottom spectral-window vector whose L^2 affine residual is at most $1/2$. The unresolved nonlinear branch can be isolated by the affine-interaction candidate

$$\text{dist}_{L^2(\mathcal{G}_\mu)}(f, \text{Aff})^2 \leq C_A \mathbb{E} \left| f \left(\frac{X+G}{\sqrt{2}} \right) - \mathbb{E}[f | X] - \mathbb{E}[f | G] \right|^2. \quad (\text{AI})$$

Its kernel is exactly affine in full dimension and its sharp Gaussian constant is two, but no universal proof is supplied here; already its quadratic sector contains a dimension-free generalized quadratic-variance theorem.

The tempting generic replacement

$$\int |w|^2 d\nu \leq C \left\{ \int \|Dw\|_{\text{HS}}^2 d\nu + \int w^T D^2V w d\nu \right\}, \quad w = \nabla f, \quad (10.3f)$$

does not provide an extra lemma. Bochner and spectral calculus show that its optimal constant is exactly $C_P(\nu)$; the centered-gradient Korn constant likewise lies between $C_P(\nu) - 1$ and $C_P(\nu)$. Thus (10.3f) is circular. The actual remaining mechanism is coherence of an almost-centered, slowly varying field which adaptively saturates the posterior covariance, or a proof of (AI) on the relevant bottom spectral windows.

9.4 Checkpoint 17: posterior saturation and affine stability

Let π be centered 1-strongly log-concave on its intrinsic Euclidean support, let $Z \sim \pi$, let $A = \text{Cov } Z$, and put $\delta_\pi(u) = |u|^2 - u^T A u$. The audited mixed-third-moment theorem is

$$M_\pi(u) := \mathbb{E}[(u \cdot Z) Z Z^T], \quad \|M_\pi(u)\|_{\text{HS}} \leq 4\sqrt{\delta_\pi(u)}. \quad (17.1)$$

For $Y = X + G$ with intrinsic $G \sim N(0, I)$, write $m_y = \mathbb{E}[X | Y = y]$, $Z_y = X - m_y$, $A_y = \text{Cov}(X | Y = y)$, and $H(y) = I - A_y$. Differentiation gives $DA_y[v] = \mathbb{E}[(v \cdot Z_y) Z_y Z_y^T | Y = y]$, and hence, for every (even adaptively selected) $u = u(y)$,

$$\left(\sum_j \|(DA_y[e_j])u\|^2 \right)^{1/2} \leq 4\sqrt{u^T H(y)u}. \quad (17.2)$$

This is a basis-summed Hilbert–Schmidt estimate with no trace or rank factor; it is local and does not synchronize directions globally.

For every locally Sobolev vector field W , the same estimate and $0 \preceq H \preceq I$ give

$$D(HW)[v] = (DH[v])W + H(DW[v]), \quad \|D(HW)\|_{\text{HS}} \leq \|DW\|_{\text{HS}} + 4\sqrt{W^T H W},$$

and therefore, with $\mathcal{D} = \mathbb{E} \|DW\|_{\text{HS}}^2$ and $\mathcal{K} = \mathbb{E}[W^T H W]$,

$$\mathbb{E} \|D(HW)\|_{\text{HS}}^2 \leq 2\mathcal{D} + 32\mathcal{K}, \quad \mathbb{E}|HW|^2 \leq \mathcal{K}. \quad (17.3)$$

The audited affine-stability lemma says that for centered 1-strongly log-concave π ,

$$D_\pi(f) := \mathbb{E}|\nabla f|^2 - \text{Var}_\pi f \geq 0 \implies \inf_{c,\ell} \mathbb{E}|f(Z) - c - \ell \cdot Z|^2 \leq 22D_\pi(f). \quad (17.4)$$

For a bottom input spectral-window vector g in the channel $Y = X + G$, put $q = \mathbb{E}|\nabla g|^2$ and $d = \mathbb{E} \text{Var}(g(X) | Y)$. The reverse comparison gives $q - d = O(a^2)$ whenever $a = \lambda_1(\mu) \downarrow 0$ (window width $o(a^2)$). Applying (17.4) in each posterior gives

$$\mathbb{E}_Y \inf_{c,\ell} \mathbb{E}_{\pi_Y} |g(X) - c - \ell \cdot (X - m_Y)|^2 \leq 22(q - d) = O(a^2). \quad (17.5)$$

The affine coefficients may switch with y ; (17.5) is not a global Gaussian-factor theorem.

For the same intrinsic channel, let $\lambda = \lambda_1(\mathcal{L}(Y))$ and let $b = 2\lambda$ be the gap of the normalized output. Spectral-window passage gives

$$\frac{\sqrt{1+12a}-1}{6} \leq \lambda \leq \frac{a}{1-a}, \quad a - 3a^2 \leq \lambda \leq a + 2a^2 \quad (a \downarrow 0),$$

and therefore

$$\lambda = a + O(a^2), \quad b = 2a + O(a^2). \quad (17.6)$$

No spectral-edge attainment is used in these comparisons.

9.5 Simple eigengap and the longitudinal frontier

Let $H = D^2U$, let $h(y)$ be its simple lowest eigenvalue, and assume $\lambda_2(H(y)) - h(y) \geq \kappa > 0$. If $P(y)$ is the rank-one lowest spectral projection, the audited Davis–Kahan/Hilbert–Schmidt calculation gives

$$\sum_j \|D_j P(y)\|_{\text{HS}}^2 \leq \frac{32}{\kappa^2} h(y). \quad (17.7)$$

Writing $Q = I - P$, the field ledger (with $\mathcal{D}, \mathcal{K}$ as above) is

$$\mathbb{E}|QW|^2 \leq \mathcal{K}/\kappa, \quad \mathbb{E}\|D(PW)\|_{\text{HS}}^2 \leq 2\mathcal{D} + \frac{32}{\kappa^2} \mathcal{K}. \quad (17.8)$$

The twice-Bochner ledger is dimension-free but leaves an additional transverse remainder. With the Witten one-form operator $\mathcal{A}_1 = -L + H$ and $C = DW = \alpha P + C_0$, $\alpha = e^T C e$, and $t = D^3U[e, e, W]$, it reads

$$\begin{aligned} \|\mathcal{A}_1 W\|_2^2 &\geq \mathbb{E}\|D(DW)\|_{\text{HS}}^2 + 3\mathbb{E}[h\alpha^2] + \frac{3\kappa}{4} \mathbb{E}\|C_0\|_{\text{HS}}^2 + 2\mathbb{E}[\alpha t] + \mathbb{E}|HW|^2 \\ &\quad - \frac{64}{3\kappa} \mathcal{K}. \end{aligned} \quad (17.9)$$

For a normalized bottom field the final term is $O_\kappa(\lambda)$, not $O_\kappa(\lambda^2)$, so it is not absorbed by (17.9). The remaining frontier therefore has both this transverse saturation remainder and the scalar longitudinal quotient gate.

If the low line is fixed, the latter gate closes by the one-dimensional Hardy factorization. For a one-dimensional log-concave $q = e^{-u}$ with $\text{Var}_q(s) \leq 2$, and $h = u''$, every smooth compactly supported amplitude a satisfies

$$\mathbb{E}(a'')^2 + 3\mathbb{E}[h(a')^2] + 2\mathbb{E}[aa'h'] + \mathbb{E}[h^2 a^2] \geq \frac{1}{24} \{\mathbb{E}(a')^2 + \mathbb{E}[ha^2]\}. \quad (17.10)$$

Thus a nonzero scalar one-form eigenmode has eigenvalue at least $1/24$, and the fixed-global-eigenline/product branch is closed. A generic amplitude-connectivity replacement is false even for exact analytic gradients: for $\nu = N(0, 2I_2)$, $f_R = (e^{Rx} + e^{Ry})/R$, and $W_R = \nabla f_R$, define $d\sigma_R = |W_R|^2 d\nu / (2e^{4R^2})$ and $N_R = W_R W_R^T / |W_R|^2$. Then

$$\inf_{\text{rank } P_0=1} \mathbb{E}_{\sigma_R} \|N_R - P_0\|_{\text{HS}}^2 = 1 - e^{-2R^2}, \quad \mathbb{E}_{\sigma_R} \|DN_R\|_{\text{HS}}^2 \leq R^2 e^{-2R^2}. \quad (17.11)$$

This example has repeated H and high field energy, so it does not disprove the simple-low-mode program; it does disprove generic amplitude-biased Poincare/connectivity claims.

9.6 Exact ANOVA jet, product closure, and matrix-Stein obstruction

For $\mu = \otimes_i \mu_i$, with every factor centered, variance-one, and log-concave, let $f \in L^2(\mathcal{L}(S))$ be centered and put $S = (X + G)/\sqrt{2}$, $h(X) = \mathbb{E}[f(S) \mid X]$, and $R = f(S) - \mathbb{E}[f \mid X] - \mathbb{E}[f \mid G]$, Gaussian Hermite jets give

$$\mathbb{E}R^2 = \sum_{|\alpha| \geq 1} \frac{\text{Var}_\mu(\partial^\alpha h)}{\alpha!}, \quad \|f\|_2^2 = V_0 + \sum_{k \geq 1} (V_k + M_k).$$

The diagonal one-dimensional Stein ledger gives the dimension-free product affine-interaction theorem

$$\text{dist}_{L^2(\mathcal{L}(S))}(f, \text{Aff})^2 \leq 900 \mathbb{E}R^2. \quad (17.12)$$

For a general centered isotropic law, a conditional frame-free criterion is the following: if a matrix Stein kernel τ satisfies $\mathbb{E}\tau = I$ and

$$K^2 = \left\| \mathbb{E}[(\tau - I)(\tau - I)^T] \right\|_{\text{op}} < \infty,$$

then, with Poincare constant C_0 , the same jet argument gives

$$\text{dist}(f, \text{Aff})^2 \leq (C_0 + 3 + 2K^2) \mathbb{E}R^2. \quad (17.13)$$

Known general log-concave kernels have only a Hilbert–Schmidt/trace budget of order n ; the operator bound in (17.13) is unresolved and is itself KLS-strength. Thus (17.12) is a product theorem, not the missing irreducible ANOVA inequality.

9.7 Why the evident reverse arguments fail

Write $\mathcal{E}_\eta(u) = \int |\nabla u|^2 d\eta$. If f is an attained normalized bottom mode of μ , define

$$H = \frac{f(X) + f(Y)}{\sqrt{2}}, \quad g = \mathbb{E}[H \mid S], \quad r = H - g.$$

Orthogonal projection and the weak eigenvalue equation give

$$\boxed{\mathcal{E}_\nu(g) = a(1 - 2\|r\|_2^2) + \mathcal{E}_{\mu^2}(r).} \quad (10.4)$$

Conditional projection need not decrease energy. For $\mu = \text{Unif}[-1, 1]$,

$$f(x) = \sqrt{2} \sin(\pi x/2), \quad a = \pi^2/4.$$

Writing $t = \sqrt{2}s$, direct averaging on the sum fiber gives

$$g(s) = \frac{4 \sin(\pi t/2)}{\pi (2 - |t|)}, \quad |t| < 2.$$

If $J = \int_0^\pi \sin^2(v) dv/v$, then

$$\text{Var}_\nu(g) = \frac{8J}{\pi^2}, \quad \mathcal{E}_\nu(g) = 4(J - \tfrac{1}{2}) > \frac{\pi^2}{4} = \mathcal{E}_{\mu^2}(H), \quad (10.5)$$

where

$$J \geq \frac{\pi^2}{8} - \frac{\pi^4}{192} + \frac{\pi^6}{13824} + \frac{5}{16} + \frac{1}{4\pi^2} > \frac{1}{2} + \frac{\pi^2}{16}. \quad (10.6)$$

Thus discarding the fiber residual cannot reverse (SC).

Several scalar Lyapunov budgets fail for independent reasons. For

$$X_k = \frac{\Gamma(k, 1) - k}{\sqrt{k}},$$

normalized self-convolution maps X_k to X_{2k} , while

$$\lambda_1(X_k) = \frac{k^2}{(k+1)^2}. \quad (10.7)$$

Raw Fisher information is $k/(k-2)$ for $k > 2$ and is infinite at the mandatory hard-support cases $k \leq 2$. Gaussian regularizations of the shifted exponential retain the limiting gap change $1/4 \mapsto 4/9$, while the decrement of every bounded inverse-Fisher transform tends to zero.

There is also a smooth high-chaos obstruction. Let H_m be the probabilists' Hermite polynomial, fix even $m \geq 4$, and variance-normalize the strongly log-concave density proportional to

$$\exp[-x^2/2 - \varepsilon H_m(x)/\sqrt{m!}].$$

For fixed m and $\varepsilon \downarrow 0$, its gap a and the gap b after normalized self-convolution satisfy

$$a = 1 - \frac{m}{m-2}\varepsilon^2 + O_m(\varepsilon^3), \quad b = 1 - \frac{m}{m-2}2^{2-m}\varepsilon^2 + O_m(\varepsilon^3). \quad (10.8)$$

For isotropic $\eta = \mathcal{L}(Z)$, define

$$\mathcal{F}(\eta) = I\left(\frac{Z+G}{\sqrt{2}} \parallel \gamma\right) = 1 - 2 \text{mmse}(Z \mid Z+G).$$

Then

$$\lim_{\varepsilon \downarrow 0} \frac{\log(b/a)}{\mathcal{F}(\mu) - \mathcal{F}(\mathbb{T}\mu)} = \frac{2^m}{m-2}. \quad (10.9)$$

Letting m increase rules out a universal fixed-noise scalar charge. Entropy survives this high-chaos test, but raw entropy and dependence deficits are additive under products, normalization by dimension fails under Gaussian padding, and bounded transforms saturate. The correct Hadamard sign is

$$I(S; D) = h(S) + h(D) - 2h(X);$$

for shifted exponentials the value is Euler's constant. These tests leave open directional, operator-valued, and function-specific reverse quantities.

9.8 The Gaussian-channel affine gate

Let $Y = X + G$, let q be the law of Y , and suppose that a smooth g satisfies

$$\mathbb{E}_\mu g = 0, \quad \mathbb{E}_\mu[Xg] = 0, \quad h(y) = \mathbb{E}[g(X) \mid Y = y].$$

The output score and affine orthogonality give

$$|\mathbb{E}_q \nabla h| \leq 2^{-1/2} \|g\|_2. \quad (10.10)$$

For the posterior μ_y , write

$$A_y = \text{Cov}_y(X), \quad a_y = \mathbb{E}_y \nabla g,$$

and let r_y be the residual after subtracting the posterior mean and best affine predictor from g . Posterior strong convexity yields

$$\mathbb{E}_\mu \nabla g - \mathbb{E}_q \nabla h = \mathbb{E}[(I - A_Y)a_Y] - \mathbb{E} \text{Cov}_Y(r_Y, X), \quad (10.11)$$

$$|\mathbb{E} \text{Cov}_Y(r_Y, X)| \leq \|D^2 g\|_{L^2(\mu)}. \quad (10.12)$$

The exact uncontrolled term is

$$\boxed{|\mathbb{E}[(I - A_Y)a_Y]| \leq C \|D^2 g\|_{L^2(\mu)}}. \quad (\text{AFF})$$

A pointwise posterior form is false, since a posterior-affine function has zero Hessian. The integrated estimate (AFF) remains unproved and is a KLS-strength near-linear gate; inserting a dimension-free Poincaré inequality for q would be circular.

9.9 The exact weighted-Fisher ledger

In the regular conditional-slice setting, let

$$r(s, z) = e^{-U(s, z)}, \quad \rho(s) = \int r(s, z) dz = e^{-\Phi(s)}, \quad q_s(z) = \frac{r(s, z)}{\rho(s)}.$$

With

$$L_s = \Delta_z - \nabla_z U(s, \cdot) \cdot \nabla_z, \quad L_s g_s = \ell_s := \partial_s \log q_s, \quad F_s = \nabla_z g_s, \quad H_s = D_{zz}^2 U,$$

put

$$C_s = \mathbb{E}_s \|D_z F_s\|_{\text{HS}}^2, \quad B_s = \mathbb{E}_s \langle H_s F_s, F_s \rangle.$$

Conditional Bochner gives

$$I_\perp(s) := \mathbb{E}_s \ell_s^2 = C_s + B_s. \quad (10.13)$$

Define

$$\alpha_s = \partial_s g_s - \frac{1}{2} |F_s|^2, \quad \mathcal{Q}_s = D^2 U[(1, -F_s), (1, -F_s)].$$

Joint convexity makes $\mathcal{Q}_s \geq 0$, and direct differentiation gives

$$L_s \alpha_s = \Phi'' - \mathcal{Q}_s - \|D_z F_s\|_{\text{HS}}^2, \quad \Phi'' = \mathbb{E}_s \mathcal{Q}_s + C_s. \quad (10.14)$$

If the scalar marginal is centered and variance one and τ is its canonical Stein kernel, then

$$\boxed{\int \rho \tau^2 (C_s + \mathbb{E}_s \mathcal{Q}_s) ds = \int \rho \tau^2 \Phi'' ds = 1 - \mathbb{E}_\rho(\tau')^2 \leq 1.} \quad (10.15)$$

Thus deformation and material curvature already have a universal budget. The conditional curvature term B_s does not: in the full-space Hodge identity its apparent contribution cancels exactly against the horizontal and mixed terms.

For the conditional centroid $m(s) = \mathbb{E}_s Z$, the same calculation gives

$$m' = -\mathbb{E}_s F_s, \quad m'' = -\mathbb{E}_s[(Z - m)(\mathcal{Q}_s + \|D_z F_s\|_{\text{HS}}^2)], \quad \mathbb{E}[\tau(S)m'(S)] = 0. \quad (10.16)$$

These global-in- s identities are indispensable: a slice-local replacement is false below.

9.10 Two complete weighted-Fisher subclasses

Pure translations. Suppose before transverse smoothing that

$$p_s(x) = Z_\varphi^{-1} e^{-\varphi(x-m(s))}$$

and that the joint law is isotropic and log-concave. If Y has density proportional to $e^{-\varphi}$, isotropy gives $\text{Cov}(Y) \preceq I$. Including subgradients and normal cones in the nonsmooth case,

$$\overline{\text{conv}}\{\partial\varphi(y)\} \supset \frac{3}{16} B_2^d. \quad (10.17)$$

Testing joint convexity against this score-range inradius yields, in distributions,

$$|Dm'| \leq \frac{16}{3} D\Phi'. \quad (10.18)$$

Cross-isotropy and one-dimensional Stein estimates give

$$\mathbb{E}[\tau^2 |m'|^2] \leq 3208 \left(\frac{16}{3}\right)^2. \quad (10.19)$$

After unit transverse Gaussian smoothing, Gaussian-channel contraction gives $I_\perp(s) \leq |m'(s)|^2$. Consequently

$$\int \rho \tau^2 I_\perp ds \leq C \quad (10.20)$$

for every pure translating family, uniformly in transverse dimension. The finite-difference proof includes asymmetric fibers and hard endpoints.

Smooth post-noise Gaussian fibers. Consider

$$p(s, z) = Z^{-1} \exp \left[-W(s) - \frac{1}{2}(z - m(s))^T Q(s)(z - m(s)) \right], \quad R(s) = Q(s)^{-1}.$$

Assume $W, m, R \in C^2$, $R \succ 0$, the joint potential is convex, the S -marginal is centered with variance one, and

$$\mathbb{E}[S m(S)] = 0, \quad I \preceq R(s) \text{ for every } s, \quad \mathbb{E}R(S) \preceq \Lambda I. \quad (10.21)$$

These are precisely the smooth full-support hypotheses used here; the lower bound on R is the unit transverse noise.

Put $y = z - m(s)$. The exact centered Poisson field is

$$F_s = -m'(s) + A_s y, \quad A_s R + R A_s = -R'.$$

Writing

$$B_s = m'(s)^T R^{-1} m'(s) + B_s^0, \quad J_s = \frac{1}{2} \text{Tr}(R^{-1} R' R^{-1} R'),$$

diagonalization of $R(s)$ at fixed s , followed by pairing the (i, j) and (j, i) terms, gives the noncommutative identity

$$\boxed{C_s + B_s^0 = J_s}. \quad (10.22)$$

No derivative of the diagonalizing frame is taken. Joint convexity gives

$$\Phi'' = W'' + \frac{1}{2} \text{Tr}(R^{-1}(-R'')) + J_s \geq J_s, \quad (10.23)$$

and hence

$$\int \rho \tau^2 (C_s + B_s^0) ds \leq 1. \quad (10.24)$$

The centroid theorem has the following audited scope: for a C^2 , full-support jointly convex Gaussian family satisfying the marginal and mixed-isotropy hypotheses above, but requiring only $\mathbb{E}R \preceq \Lambda I$, matrix concavity and the Schur condition imply

$$\mathbb{E}[\tau^2 |m'|^2] \leq C_{\text{cent}} \Lambda. \quad (10.25)$$

This allows noncommuting, rotating covariance matrices. Under the post-noise lower bound $R \succeq I$, one has $R^{-1} \preceq I$, so

$$\boxed{\int \rho(s) \tau(s)^2 (C_s + B_s) ds \leq 1 + C_{\text{cent}} \Lambda.} \quad (10.26)$$

No nonsmooth Gaussian limit and no zero-noise limit is asserted. A separate hard-interval calculation controls geometric centroid velocity, including wedge curvature atoms, but its identification with the post-noise Poisson field has not been proved.

9.11 A cone forcing global Stein weighting

For $a \geq 2$, let

$$q_a = \text{Unif}[-a, a] * \gamma, \quad U_a = -\log q_a.$$

The mean-zero gradient field $F_a(x) = x/a$ satisfies

$$\mathbb{E}_{q_a} |F'_a|^2 = a^{-2}, \quad \mathbb{E}_{q_a} [U''_a F_a^2] \geq \frac{1}{64a}. \quad (10.27)$$

More decisively, the same divergence occurs for the exact Poisson field of one fixed isotropic law. Start from

$$p(r, x) = \frac{1}{2} e^{-r} \mathbf{1}_{\{r>0, |x|<r\}}, \quad S = \frac{R-2}{\sqrt{2}}, \quad Y = \frac{X}{\sqrt{2}}.$$

Then $Y \mid S = s$ is uniform on $[-a, a]$, where $a = s + \sqrt{2} > 0$. After unit Gaussian smoothing in Y , its exact centered Poisson field obeys

$$\mathbb{E}_s F_s = 0, \quad C_s \leq a^{-2}, \quad B_s \geq \frac{1}{256a}. \quad (10.28)$$

Thus no universal slice-by-slice estimate $B_s \lesssim C_s + |\mathbb{E}_s F_s|^2$ can hold. Nevertheless

$$\rho(s) = 2ae^{-\sqrt{2}s} \mathbf{1}_{\{s>0\}}, \quad \tau(s) = \frac{a}{\sqrt{2}}, \quad B_s \leq 1,$$

and therefore

$$\int \rho \tau^2 B_s ds \leq \frac{3}{2}. \quad (10.29)$$

The obstruction is entirely in the centered shape residual, not centroid motion. It forces a proof to retain the joint s -geometry and Stein weight.

9.12 Exact remaining gates

The weighted route would close its general slice estimate if one proved

$$\boxed{\int \rho \tau^2 B_s ds \leq C \int \rho \tau^2 (C_s + \mathbb{E}_s Q_s) ds.} \quad (\text{WK})$$

The cone proves that (WK) must be integrated in s , while the translation theorem removes pure centroid motion from the unresolved sector. What remains is genuinely nonlinear shape change. A proposed scalar shape-acceleration shortcut is invalid: midpoint expansion cancels the $U_z m''$ term and yields only affine-path curvature.

The two surviving near-linear gates are (AFF) and (WK), but even a full weighted-Fisher or directional-MMSE theorem would currently close only the near-linear spectral branch. The tensor-extremizer program still lacks a proved dimension-free dichotomy forcing every hypothetical small-gap mode to be near-linear, or an alternative argument applying the weighted estimate to every low mode. Neither (AFF), (WK), nor this global nonlinear/near-linear dichotomy is assumed here.

10 Mechanisms ruled out by explicit tests

The negative results are as useful as the positive ones because they prevent equivalent forms of the conjecture from being reintroduced under new notation.

Proposed mechanism	Audited obstruction
Thin shell alone controls (TM)	Projection traces permit the harmonic spectrum $j^{-1/2}$, leaving $\sqrt{\log n}$.
Full-gradient interpolation	The first Neumann mode of the isotropic ball violates the natural multiplicative interpolation by $\sqrt{n}$.
Reverse conditional projection	For the first interval mode, $\mathcal{E}_\nu(\mathbb{E}[H S]) > \mathcal{E}_{\mu^2}(H)$; the exact residual identity (10.4) has no favorable sign.
Scalar reverse-convolution Lyapunov	Hard-support Gamma laws defeat bounded inverse-Fisher budgets, high Hermite modes defeat fixed-noise Fisher/MMSE charging, and additive entropy budgets fail under products or Gaussian padding.
Trace localization $\Rightarrow$ selected direction	A one-spike product-integral path has $O(n)$ total trace but polylogarithmic amplification in the chosen direction.
Coarea or determinant control of (FA)	The flow expectation is at a fixed initial tilt. A linear map $\text{diag}(\sqrt{n}, 1, \dots, 1)$ has acceptable trace and determinant scale but normal stretch n .

Proposed mechanism	Audited obstruction
Strong self-concordance $\Rightarrow$ KLS	On an isotropic cube, an explicit strongly self-concordant barrier has weighted Poincaré constant of order n . Barrier axioms give no covariance comparison.
Bakry–Émery for the canonical metric	For the direct Cheng–Yau potential the natural curvature has the wrong sign; for uniform measure the curvature-to-metric ratio already tends to zero at an interval boundary.
Pointwise barrier-metric comparison	It fails on the simplex, skew cones, and product exponentials; large inverse metric can occur on rare sets.
Tensor multiplicity	Exact splitting never exceeds the convex buffer, and the low modes remain inside the allowed small-rank localization tail.
Interior Brascamp–Lieb for slices	Uniform intervals and centered exponentials carry covariance motion entirely through moving-boundary terms.
Pointwise post-noise curvature–Korn	The exact cone Poisson field has $C_s \leq a^{-2}$, $B_s \geq (256a)^{-1}$, and zero mean, although its Stein-weighted integral is at most $3/2$.
Scalar shape acceleration	Midpoint expansion cancels the proposed $U_z m''$ term and leaves only affine-path curvature.
Finite Gaussian chaos	Exponential examples have factorial mixed moments; the relevant Hermite expansion can have zero radius of convergence.
Entropy-controlled convex certificate	A one-dimensional zero-deficit transport defeats every prescribed-orientation signed convex certificate.

11 Prioritized future directions

The following program is ordered by how sharply each step is formulated and how independently it can be tested.

11.1 Priority I: posterior covariance and low-mode alignment

1. Prove or disprove (DMMSE). First establish the proposed dimension-descent estimate (7.9) with all conditional derivatives and tail bounds written rigorously. Then identify an additional rigidity input that removes its logarithm. Candidate inputs include a tail theorem for the rank-one survivor variance or a nonlinear-information bound forced by joint log-concavity.
2. Attack (FA) directly as a matrix-martingale statement. Any proof must use that c_t is a posterior gradient of a fixed scalar low mode; trace bounds and arbitrary terminal directions are insufficient. A useful intermediate target is a bound on amplification-weighted occupation of $(A_t - 3I)_+$ along the direction selected by c_t .
3. Relate the two targets. The posterior split (7.2) shows that (DMMSE) closes very-near-linear modes, while (FA) is a pathwise refinement. A proof should quantify precisely how much residual size δ each one tolerates.
4. Prove or disprove (AFF). Equations (10.11)–(10.12) already control every posterior-nonlinear

component; any proof must control the remaining posterior-affine term globally rather than by a false pointwise Hessian estimate.

11.2 Priority II: non-Gaussian slice-curvature compensation

The smooth post-noise Gaussian theorem proves the full weighted-Fisher bound by the exact identity $C_s + B_s^0 = J_s$, even for noncommuting covariance motion. The next target is the integrated estimate (WK) for general slices. The cone calculation forbids replacing it by a slice-by-slice Korn inequality, while the translation theorem shows that pure centroid motion is no longer the issue.

One possible route to (WK) is a boundary-inclusive version of the Brascamp–Lieb stability estimate

$$\left\| C^{-1/2} \text{Cov}_t(g, (Y - m)(Y - m)^T) C^{-1/2} \right\|_{\text{HS}}^2 \lesssim \mathcal{D}_{\text{BL}}(g) + \text{boundary/Reilly charge.} \quad (9.1)$$

The endpoint term must reproduce the interval and exponential examples exactly. Before attempting full generality, test (9.1) on polytopal slices with rotating facet normals and on smooth approximations of boxes.

The affine-Hessian inequality (AH) is a clean alternative formulation. It should first be proved or falsified for centrally symmetric but dependent bodies, uniform polytopes, and skew cones. Any proof for all gradients would contain a KLS-strength near-linear statement, so a score-specific version may be more realistic.

11.3 Priority III: exact canonical metric, only for low modes

Strong self-concordance alone is blocked. A narrower question remains: for the exact direct Cheng–Yau metric $H = D^2 F$, can a first Euclidean eigenfunction satisfy

$$\int \langle H^{-1} \nabla f, \nabla f \rangle d\mu \leq C \left(\int |\nabla f|^2 d\mu + \int \|D^2 f\|_{\text{HS}}^2 d\mu \right)? \quad (9.2)$$

Bochner supplies the second term at order λ^2 . The central issue is whether a low mode can concentrate its gradient on the rare sets where H^{-1} is large. The simplex and skew-cone calculations provide sharp stress tests. This direction is worthwhile only if it uses the exact canonical equation; generic barrier axioms have already been ruled out.

11.4 Priority IV: extremal structure

If one pursues an extremal sequence, impose all known constraints at once:

- witnesses are nonlinear and non-radial;
- tensor-product explanations are factored out;
- near-linear witnesses must satisfy the posterior and localization constraints (6.5), (FA), and (7.5);
- any tensor stationarity argument must pay the exact convex-buffer cost (8.3);
- any slicing argument must retain moving-boundary curvature.

A useful structure theorem would state that every counterexample sequence has one of two mutually exclusive features: a direction with vanishing unit-noise posterior MMSE, or a genuinely high-rank

noncommuting conditional-covariance motion. The two active programs above are designed to exclude precisely those alternatives.

12 Audit requirements for any future proof

Before a candidate argument is regarded as a proof, it should pass the following checks.

1. Track n , smoothing, truncation, localization time, stopping rank, and bootstrap depth symbolically. The final constant must be independent of all of them.
2. Verify stochastic existence, stopping, integrability, and true martingale properties rather than formal local-martingale calculations.
3. Pass from smooth, compact, strongly log-concave approximations to arbitrary log-concave measures without changing the constant.
4. Treat lower-dimensional affine supports intrinsically and exclude only point masses.
5. Cover every locally Lipschitz L^2 test function in a Poincaré proof.
6. Scan for circular assumptions: uniform Poincaré, (M), (TM), (MG), dimension-free survivor covariance, (AFF), (WK), and strong self-concordance of the entropic barrier are all KLS-strength gates unless independently proved.
7. Test the constant chain on the cube, regular simplex, crosspolytope, product exponential, Euclidean ball, and a skew non-symmetric cone.
8. Have a clean-room reader reprove every load-bearing lemma from its statement alone.

13 Conclusion

The investigation has not resolved KLS. The strict convolution inequality (SC), including the unequal-input form (USC), remains a forward amplification statement. The independent posterior comparison

$$a \geq \frac{b}{2+b}$$

does give a dimension-free reduction to positive analytic isotropic outputs with $0 \preceq D^2 \tilde{U} \preceq 2I$. Along a hypothetical small-gap sequence this comparison is now known to saturate to second order: the unscaled output gap is $a + O(a^2)$, and bottom input modes saturate the posterior Poincaré and covariance inequalities on average.

Checkpoint 17 strengthens the local rigidity picture without closing its global gate. The Hilbert–Schmidt posterior theorem removes the former trace loss, and strongly-log-concave affine stability shows that bottom modes are posterior-affine on average up to squared error $O(a^2)$. Under a uniform simple posterior eigengap, the low spectral projection has dimension-free derivative energy. Nevertheless the twice-Bochner ledger still contains an $O_\kappa(\lambda)$ transverse saturation remainder and a signed longitudinal quotient term. The longitudinal term is coercive with constant $1/24$ after the line is fixed, but explicit Gaussian examples rule out generic amplitude-weighted connectivity as a way to fix it.

The exact Gaussian-jet identity also sharpens the nonlinear branch. It proves the affine-interaction estimate with constant 900 for arbitrary products of one-dimensional isotropic log-concave laws.

Extending that argument to irreducible laws would require a new operator-level matrix-Stein or equivalent coercivity mechanism; known trace budgets and the input Poincaré inequality are insufficient.

Accordingly, the current frontier has two coupled components: global posterior direction and amplitude synchronization strong enough to absorb simple-eigenspace remainders, and a dimension-free affine-interaction or nonlinear/near-linear dichotomy for general log-concave laws. The older weighted-Fisher gates (AFF), (WK), (DMMSE), and (FA) remain alternative formulations, not assumptions. None of the audited reductions, stability theorems, fixed-line arguments, or subclass closures in this note is a complete proof of KLS.

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